

Saturation Performance of IEEE 802.11 Broadcast Networks

Xiaomin Ma, *Member, IEEE*, and Xianbo Chen, *Student Member, IEEE*

Abstract—In this letter, saturation performance of broadcast service in IEEE 802.11 is studied analytically and by simulation. Having investigated the features of the broadcast service, we point out that the analytic models for saturation performance evaluation of IEEE 802.11 unicast communication cannot be simply reduced for analysis of broadcast service. Under realistic assumptions, we construct an analytic model to characterize operation of the backoff counter for broadcast, thus obtain closed form solutions to the saturation throughput and packet delivery ratio. Simulation validates the proposed model. Numerical results reveal characteristics of the broadcast service.

Index Terms—IEEE 802.11, broadcast, medium access control, throughput, performance evaluation.

I. INTRODUCTION

ONE of the most popular MAC protocols for ad hoc wireless networks is CSMA/CA (Carrier sense multiple access with collision avoidance) with binary exponential backoff used by IEEE 802.11 [1]. The saturation performance of the distributed coordination function (DCF) mode of IEEE 802.11 MAC protocol is extensively studied [2]–[5]. However, to the best of the authors' knowledge, there has been no analytic model for performance evaluation of broadcast service in IEEE 802.11. In fact, broadcast services are provided widely in various ad hoc network applications.

In IEEE 802.11 [1], the basic access procedure is adopted for broadcast service. However, since there is no retransmission attempt or Acknowledgement (ACK) in broadcast service, the transmitting station does not know if its transmission is successful or not, and the transmitting station just assumes the sent packet has arrived at destination and sense channel again if it has more packets to transmit. So, the current backoff window size is always equal to the initial minimum backoff window size W_0 .

In the light of IEEE 802.11 standards [1], we know that transmission shall commence whenever the backoff time counter reaches zero. So, it is possible that a station that just completes a transmission and has a new packet to send starts to transmit right after a distributed inter-frame space (DIFS) if the station happens to choose zero as its initial backoff time. Such consecutive transmissions shall make other listening stations with nonzero backoff times have no chances to backoff, which is referred to as backoff counter *consecutive freeze process* (CFP). This phenomenon in unicast service has been addressed and analyzed in [3], [5]. In unicast service, after a busy period,

only station which just has a successful transmission may consecutively access the channel if its newly chosen backoff time is zero [3]. However, in broadcast service without ACK or retransmission mechanism, all stations that just finish their transmissions may have chance to consecutively access the channel even if a collision just occurs. Therefore, CFP in broadcast happens more often than that in unicast service.

In this letter, we construct a simple but very accurate model to characterize backoff counter operation in IEEE 802.11, upon which closed form expressions of performance for broadcast are derived. Results of the analytical model are validated by simulation.

II. PERFORMANCE ANALYSIS

We use the similar assumption in paper [2] about saturation conditions. In saturation conditions, each station has immediately a packet available for transmission, after the completion of each transmission. Knowing that backoff window size for each station in broadcast service keeps constant, the backoff counter can be characterized by a one-dimensional random process. However, analytic models for unicast communication [2]–[5] cannot be reduced directly to one-dimensional Markov chain for analysis of broadcast service in IEEE 802.11 because the CFP in broadcast is different from that in unicast. As a matter of fact, there is the possibility that multiple packet transmissions may take place consecutively between beginnings of two adjacent slot times (the period is referred to as a virtual slot [4]), and the backoff counter may be stopped by the CFP. However, if the backoff counter consecutive freeze scenario is taken into account, the probability that a station transmits a packet in a slot is no longer independent of backoff procedure. So, in this situation, the backoff counter can not be characterized as a Markov process. By exploiting the special features of backoff counter in IEEE 802.11 broadcast, we divide backoff counter process into two involving subprocesses: sequential backoff process (SBP) which involves general backoff procedure without zero initial backoff time and CFP which involves consecutive transmissions as a result of zero initial backoff time.

First, we construct an analytic model for the SBP in which the probability that a station transmits in a virtual slot time is assumed reasonably to be constant as it was in [2], [4]. In SBP, since zero initial backoff time is excluded and each station which has packet to send must backoff by an idle slot before any transmission, the maximum backoff value is $W_0 - 2$ [4]. Consider a fixed number n of contending stations. As described in [4], the stochastic process $\{b_k\}$, indexed by virtual slot number k , is a one dimensional discrete-time Markov chain. The state transition diagram describing backoff counter decrement is shown in Fig. 1, with the only nonnull

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X. Ma is with the Department of Engineering & Physics, Oral Roberts University, Tulsa, OK 74133, USA (e-mail: xma@oru.edu).

X. Chen is with the School of Electrical and Computer Engineering, University of Oklahoma, Norman, OK 73019, USA (e-mail: xianbo@ou.edu).

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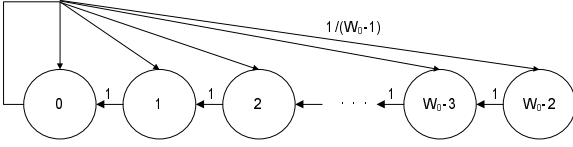


Fig. 1. Markov chain model for SBP in broadcast

one-step transition probabilities being

$$\begin{cases} P\{k|k+1\} = 1, & k \in [0, W_0 - 3]; \\ P\{k|0\} = 1/(W_0 - 1), & k \in [1, W_0 - 2]. \end{cases}$$

From Fig. 1, we can derive following relations through chain regularities

$$b_k = \frac{W_0 - k - 1}{W_0 - 1} b_0, \quad 0 < k \leq W_0 - 2 \quad (1)$$

$$\sum_{k=0}^{W_0-2} b_k = 1. \quad (2)$$

Let τ_s be the probability that a station transmits in the SBP during a virtual slot time. As any transmission occurs when the backoff counter value is equal to zero, solving (1) and (2), we have

$$\tau_s = b_0 = \frac{2}{W_0}. \quad (3)$$

Then, we analyze the CFP. Packet transmissions triggered by zero initial backoff time could happen right after a busy period resulted from the SBP, which is referred to as first freeze stage, or could happen right after a freeze stage, which is referred to as the i th freeze stage ($i=2, 3, \dots$). Under saturation condition in broadcast, a station that just completes a transmission in the previous freeze stage will attempt to send one more packet out. It chooses zero as its initial backoff time with a uniform probability $1/W_0$. The CFP will continue until zero initial backoff time is no longer chosen by any station which attempts to send new packets out. Here, we try to approach performance analysis of the CFP through evaluating transmission probability of each station in the freeze stages. Define $\tau_f(i)$ ($i=1, 2, \dots$) to be the probability that a station transmits i packets consecutively in a virtual slot due to zero initial backoff time. Hence, given the initial probability τ_s , the probability that such a station transmits in the first stage of the freeze process are derived as

$$\tau_f(1) = \frac{\tau_s}{W_0}.$$

Consequently, the probabilities that such a station transmits in the i th stage of the freeze process can be evaluated as

$$\tau_f(i) = \frac{\tau_f(i-1)}{W_0} = \frac{\tau_s}{W_0^i}, \quad i = 2, 3, \dots \quad (4)$$

Now, we consider evaluation of channel performance resulted from SBP. Let p_{bs} denote the probability that the channel is busy as a result of SBP. Given τ_s , we have

$$p_{bs} = 1 - (1 - \tau_s)^n. \quad (5)$$

Furthermore, the probability p_{ss} that a successful transmission occurs in a slot time is derived as

$$p_{ss} = 1 - (1 - \tau_s)^{n-1}. \quad (6)$$

Then, we consider the channel performance resulted from the CFP. Knowing $\tau_f(i)$, $i = 1, 2, \dots$, the probability $p_s^f(i)$ that a successful transmission occurs in the i th freeze stage is obtained

$$p_s^f(i) = n\tau_f(i)(1 - \tau_f(i))^{n-1}. \quad (7)$$

Similarly, the probability $p_b^f(i)$ that the channel is busy in the i th freeze stage, is derived as

$$p_b^f(i) = 1 - (1 - \tau_f(i))^n. \quad (8)$$

Let $E[N_{sf}]$ denote the average number of successful transmissions in the CFP in a virtual slot time, which is expressed as

$$E[N_{sf}] = \sum_{i=1}^{\infty} 1 \cdot p_s^f(i) = \sum_{i=1}^{\infty} \frac{n\tau_s}{W_0^i} (1 - \frac{\tau_s}{W_0^i})^{n-1}. \quad (9)$$

Under the conditions $\frac{\tau_s}{W_0^i} = \frac{2}{W_0^{i+1}} \ll 1$, $i > 0$, and $W_0^i \gg n > 1$,

$$(1 - \frac{\tau_s}{W_0^i})^n \approx 1 - \frac{n\tau_s}{W_0^i}$$

holds, and provides the following approximate expression:

$$E[N_{sf}] \approx \frac{n\tau_s}{W_0 - 1} - \frac{n(n-1)\tau_s^2}{W_0^2 - 1}. \quad (10)$$

Let $E[N_{bf}]$ be the average number of freeze stages on which the channel is busy. We have

$$E[N_{bf}] = \sum_{i=1}^{\infty} 1 \cdot p_b^f(i) = \sum_{i=1}^{\infty} (1 - (1 - \tau_f(i))^n) \approx \frac{n\tau_s}{W_0 - 1}. \quad (11)$$

Let T_s be the average time the channel is sensed busy because of a successful transmission, T_c be the average time the channel is sensed busy by each station during a collision, and σ be the slot time duration. It is assumed that a packet holds size with average packet length $E[P]$, and packet header includes physical layer header plus MAC layer header: $T_H = PHY_{hdr} + MAC_{hdr}$. Let δ be the propagation delay, and $DIFS$ be the time period for a distributed inter-frame space.

$$T = T_s = T_c = T_H + E[P] + DIFS + \delta$$

a) *Saturation Throughput*. Let S be the normalized saturation throughput, expressed as the fraction of time the channel is used to transmit payload bits successfully in a virtual slot time. Effects on the throughput from both the SBP and the CFP are combined. Considering that there must be an idle back off slot in each virtual slot, we get

$$S = \frac{E[P]'}{\sigma + T_b'} \quad (12)$$

where $E[P]' = (1 \cdot p_{ss} + E[N_{sf}])E[P]$ is the average amount of payload bits transmitted successfully in a virtual slot; $T_b' = (1 \cdot p_{bs} + E[N_{bf}])T$ is the average time the channel is sensed busy in a virtual slot.

b) *Packet delivery ratio (PDR)* is defined as the ratio of the number of packets successfully received to the number of packets transmitted. So, PDR can be interpreted as $1 \cdot p_{ss} + E[N_{sf}]$ that the number of successfully transmitted packets

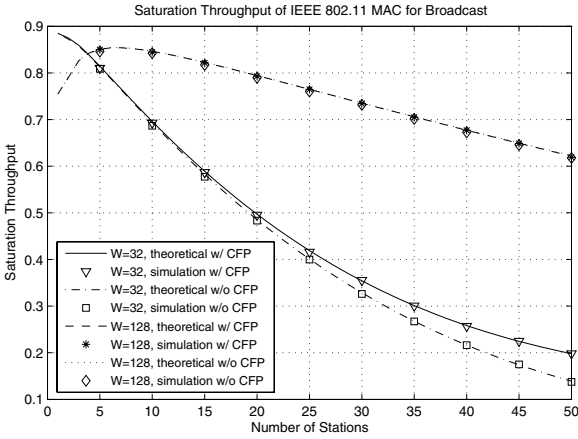


Fig. 2. Saturation throughput of IEEE 802.11 for broadcast

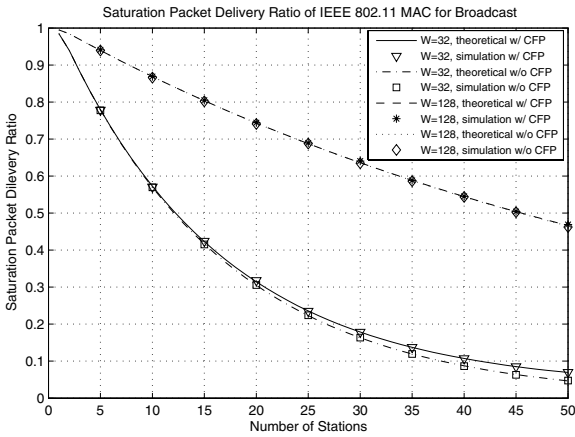


Fig. 3. Saturation packet delivery ratio of IEEE 802.11 for broadcast

during a virtual slot over $E[N_{pktvs}]$ which is the average number of stations transmitting packets in a virtual slot.

$$PDR = \frac{1 \cdot p_{ss} + E[N_{sf}]}{E[N_{pktvs}]} \quad (13)$$

Given the probability τ that a station transmits in a virtual slot, the average number of stations transmitting packets in a virtual slot is $n\tau$ [6]. So, we have

$$E[N_{pktvs}] = n\tau_s + \sum_{i=1}^{\infty} n\tau_f(i). \quad (14)$$

Substituting (3), (4), (6), (10), and (14) into (13), we obtain

$$PDR \approx \frac{(W_0 - 1)(W_0 - 2)^{n-1}}{W_0^n} + \frac{W_0 - 2n + 3}{W_0(W_0 + 1)}. \quad (15)$$

III. NUMERICAL RESULTS AND DISCUSSIONS

All the parameters used in analytical model and our simulations follow the parameters in paper [2], which are $P = 8184$ bits, $T_H = 272 + 128$ bits, system channel bit rate is 1Mbit/s, $\delta = 1\mu s$, $DIFS = 128\mu s$, and slot time $\sigma = 50\mu s$.

Fig. 2 and Fig. 3 show the saturation throughputs and PDRs over the number of stations with different backoff window sizes. As we see from Fig. 2 and Fig. 3 that the analytical model is highly accurate: analytical results (lines) practically coincide with the simulation results (symbols). We also observe that the saturation throughput for broadcast is much lower than that of basic access mechanism for unicast service in IEEE 802.11 [2] as n is greater than 5. Increasing window size helps improve the throughput and the PDR performance of broadcast services. It is observed that the throughput and PDR derived from our proposed model are higher than their counterparts obtained from the model without considering the CFP as $W_0 = 32$, $n > 20$ in Fig. 2 ~ 3. However, as bigger window size is chosen ($W_0 = 128$ in Fig. 2 ~ 3), no significant difference can be seen between performance obtained from our proposed model and that from the analytic model without taking the CFP into consideration. So, it is concluded that effect of the CFP can not be neglected in broadcast performance evaluation as the backoff window size is relatively small comparing with the number of stations in the system. Furthermore, we observe that PDR only depends on the number of stations n and selected contention window size W_0 .

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